Alarm Webs: A Framework for Decoding RAN Alarm Dynamics ⋆

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Abstract: This paper introduces "Alarm Webs", a pioneering framework designed to decode the dynamics of Radio Access Network (RAN) alarms, utilizing a novel graph-theoretic approach. Leveraging directed graphs (digraphs), this work uniquely encodes RAN alarm data, with each alarm represented as a node and its severity indicated by distinct colors. The edges of these graphs signify the sequence of alarms, creating a directional flow from one alarm to the next. Utilizing a comprehensive dataset from a major telecommunications service provider in the UK, encompassing alarms from 4G and 5G networks across over 7,000 base stations, the framework offers an unprecedented analysis of alarm patterns and interactions. The proposed methodology involves the calculation of in-degrees and out-degrees for each node, identifying those with in-degrees higher than four as significant indicators of commonly resulting alarms. This threshold-based selection is instrumental in isolating the most impactful alarms. Additionally, the application of Sankey diagrams brings an intuitive visualization of alarm causality, based on the in-degree centrality of the nodes. This approach not only clarifies the complex interrelationships among alarms but also aids in identifying potential cascading issues within the network. By effectively mapping alarm sequences and identifying key alarms, this framework paves the way for enhanced network reliability and proactive maintenance strategies in the ever-evolving telecommunications landscape.

1. INTRODUCTION

The evolution of Radio Access Network (RAN) has led to several architectures, primarily differentiated by their level of centralization and the technology used. Traditional RANs, often referred to as Distributed RANs (D-RANs), have their equipment, like base stations, distributed geographically. Each base station is self-contained with its own Remote Radio Unit (RRU), Baseband Unit (BBU), and other processing capabilities. Fig. 1 shows a typical D-RAN architecture. The RRU in a RAN converts and transmits radio signals near antennas for better signal strength. The BBU processes signals and manages network resources, primarily between the RRU and the core network. With the advent of newer technologies, such as 4G and 5G, RAN architecture has evolved into more centralized forms – Centralized RAN (C-RAN) – where the base station's baseband processing is centralized, reducing the cost and power consumption at individual sites, allowing for more efficient management of resources and better coordination between cells. RAN base stations are critical components in modern telecommunications infrastructure, facilitating wireless communication between mobile devices and a network's core. As the backbone of mobile communication, RAN base stations are equipped with sophisticated alarm systems designed to monitor and report various operational anomalies and performance issues. These alarms serve as an early warning system, enabling security alerts, timely maintenance and ensuring uninterrupted service quality (Holmbacka et al., 2018).

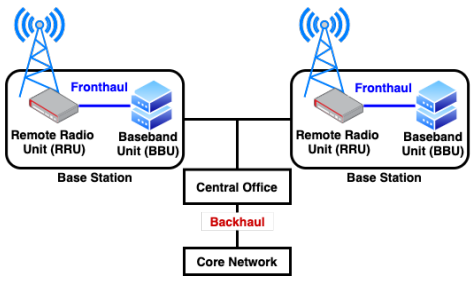

Fig. 1. A typical distributed RAN (D-RAN) architecture

Alarm systems in RAN base stations are intricate, often generating alerts that signal everything from minor glitches to major system failures. These include hardware malfunctions, software errors, connectivity issues, and environmental factors impacting the station's performance (Polese et al., 2023). The complexity of these systems arises from the variety of potential issues they monitor

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and the interdependencies between different alarms. Understanding these interdependencies is crucial for effective network management and maintenance (Awad and Hamdoun, 2016). However, deciphering the intricate web of alarm interdependencies poses a significant challenge. A single issue can trigger a cascade of alarms, making it difficult to identify the root cause of a problem. This complexity can lead to delayed response times and increased maintenance costs, ultimately affecting the quality of service experienced by end-users (Strasser and Sheppard, 2011). Further, networks have also been dependent on developing vast amounts of specialist knowledge to understand the interdependencies, and this knowledge has not always been recorded in a systematic way and is not easily available. By modeling the network of alarms as a graph, identifying patterns, isolating critical issues, and predicting potential system failures becomes easier, thereby leading to more efficient and proactive maintenance strategies (Fournier-Viger et al., 2020).

"Alarm Webs" uses graphs and robust visualization tools to develop a framework for understanding the intricacies of alarm interdependencies by asking the following research questions:

RQ1: How do graphs enhance the identification and understanding of causality in alarm systems across different base stations, and what insights does the degree centrality of nodes provide regarding the most influential alarms in frequency and severity?

RQ2: How does the visualization of alarm sequences contribute to the interpretation of complex alarm data, and how can this methodology be used to predict and prevent cascading failures within communication base stations?

Towards addressing the above research questions, this work makes the following unique contributions to the state-of-the-art:

(1) Introduces a novel approach to alarm system analytics by using digraphs to encode and analyze the relationship between RAN alarms, providing a unique method for representing and understanding the sequence and impact of different alarms.
(2) Develops a new criterion for assessing the significance of alarms in a network using in-degree centrality, enabling the identification of the most commonly resulting alarms that may indicate critical points for system vulnerability or recurrent issues.
(3) Pioneers the use of Sankey diagrams in the context of alarm sequence analysis, offering a powerful visualization tool that elucidates the flow and interdependencies of alarms, highlighting potential causal links. By filtering and analyzing nodes with high in-degree centrality, the research provides insights with the potential to predict and prevent cascading failures, contributing to the proactive maintenance and enhanced reliability of communication base stations.

2. RELATED WORK

Effective telecommunications infrastructure management hinges on the pivotal role of network alarm monitoring. The growing complexity of mobile networks has accentuated the importance of monitoring alarms within the Radio Access Network (RAN). The literature extensively explores diverse methodologies, aiming to propel the field of alarm management within telecommunications networks. Understanding the dynamics of RAN alarms is crucial for proficient fault management. Within this context, the prevalent techniques for uncovering alarm correlation and conducting root cause analysis prominently involve pattern mining and graph-based approaches. Mazdziarz (2018) employs a topology-wise k-means iterative clustering approach to identify alarm correlation clusters, demonstrating its effectiveness in enhancing response time during network fault troubleshooting. In a distinct approach, Wang et al. (2017) introduces an alarm behaviour analysis and discovery system that leverages minimal flapping rules and parent-child rules derived from extensive statistical alarm analysis to filter out trivial alarms, thereby enriching fault management capabilities. Li and Li (2010) contribute to the literature utilising a neural network-based weighted association rules mining technique. On a parallel front, Solmaz et al. (2017) incorporate sliding window-based rule mining and a graph-based approach for real-time alarm management system root-cause analyses. The application of Bayesian networks for alarm correlation estimation and prediction of un-occurred alarms is demonstrated by Hata et al. (2022), showcasing the versatility of probabilistic models in this domain. The evolving landscape of predicting RAN alarms has sparked recent research endeavours. Wang and Liang (2020) introduce a predictive analytics model for network alarms, integrating Long Short-Term Memory (LSTM) and association algorithms to forecast potential issues proactively. Massaro et al. (2023) contribute a tailored predictive model for RAN alarms, incorporating survival probability models and graph-matching techniques on historical data to anticipate potential network hardware faults. Boldt et al. (2021) expand this domain by highlighting the efficacy of Random Forest models for predicting alarms in cellular base stations, offering valuable insights for developing early warning systems. Murphy et al. (2023) further enrich the literature through a comprehensive survey, providing an overview of various machine-learning techniques for predicting faults in heterogeneous telecommunication networks.

The proposal presented here finds its closest precedents in applications rooted in graph theory. Graph theory is a powerful tool for modelling and analysing complex relationships within network alarm data. Notably, Fournier-Viger et al. (2020) introduced a dynamic graph-based approach that amalgamates historical alarm data with network topology information, providing a comprehensive framework for alarm correlation pattern analysis in telecommunication networks. This contribution exemplifies the potential of dynamic graph representations in capturing the evolving nature of alarm relationships over time. In a parallel vein, Zhang et al. (2020) proposed an innovative approach involving a hybrid causal graph learning method (HPCI) and a network embedding-based algorithm (CPBE). This methodological blend is designed to infer network weights and construct an alarm influence graph, facilitating more intricate and accurate root cause analyses. The integration of causal graph learning and network embedding showcases a commitment to refining the precision of graph-based models in the context of alarm

management. Extending the graph-based paradigm, Jiang and Bai (2023) make a significant contribution with their data-driven propagation-based fault detection approach (APGNN). This method employs alarm data mining to construct a Bayesian network, allowing for identifying root alarms and extracting a root-derived graph. This synthesis of data-driven propagation and Bayesian network construction represents a noteworthy stride toward more holistic fault detection and root cause analysis within the realm of network alarm data.

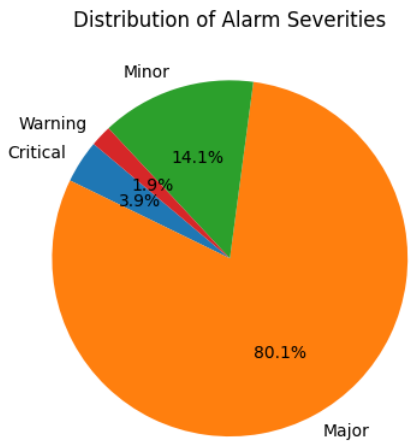


Fig. 2. Distribution of the alarm severities in the RAN dataset

In summary, the related work on alarm analytics spans real-time detection, dynamic analysis, predictive modelling, and graph-based approaches, forming a robust foundation for understanding and managing alarms in telecommunications networks. This paper advances the graph-based approach, offering novel insights into the complex dynamics of Radio Access Network (RAN) alarms by introducing the "Alarm Webs" framework, contributing to the evolving field of RAN alarm management.

3. RAN ALARM DATASET

RAN alarms are critical indicators in the telecommunications industry, providing essential insights into network infrastructures' operational health and performance. The dataset used in this work integrates RAN base station alarm data obtained from the data collection engine of British Telecom (BT) Group's alarm management system. The alarm data includes a subset of alarms from 4G and 5G network equipment generated by over 7000 selected base stations across the U.K. All the alarm data were collected between 2019 and 2021 and contained around 13.3 million alarm records. These alarms range from critical to minor and are pivotal in preemptive maintenance and troubleshooting, ensuring network reliability and efficiency. Fig. 2 shows the distribution of alarm criticality in the dataset records.

3.1 Dataset Description

Each alarm record consists of 13 attributes. Attributes 1, 2, 5, and 6 capture alarm-related details, documenting the alarms' type, importance, timestamp, and location information. Attributes 3, 4, and 13 record the base station's ID, type, and maintenance status that triggered the alarm. The remaining attributes contain operator-related information. The data formats are highly heterogeneous, consisting of integers, text, symbols, and sometimes their combinations. Mainly, Attribute 6 represents a mixed data format in the form of indefinite length and structure, resembling log entries.

Each alarm is associated with a unique attribute that aids in identifying and characterizing the alarm within the system. The severity of an alarm (Attribute 1) indicates the urgency with which it needs to be addressed. The type of alarm (Attribute 2) provides a classification that can be used to discern patterns or commonalities among the alarms. Each alarm is sourced to a particular base station, identified by an ID (Attribute 3), which allows for spatial analysis of alarm occurrences. Additional information about the alarm's source can include the base station type (Attribute 4), which can be instrumental in diagnosing systemic issues. The location of the alarm (Attribute 5) is logged for further granularity, offering insights into the precise area within the system where the issue has arisen. Timestamps are crucial for chronological analysis, capturing the moment an alarm occurs (Attribute 6), is cleared (Attribute 7), and is acknowledged (Attribute 8). The status fields indicate whether an alarm has been cleared (Attribute 9) or acknowledged (Attribute 10), which is vital for tracking the response and resolution times. Each alarm is also assigned a serial number associated with the alarm log (Attribute 11) and the equipment involved (Attribute 12), ensuring a thorough audit trail. Finally, the maintenance status of the base station (Attribute 13) can offer context as to whether an alarm might result from ongoing issues or maintenance activities. Collectively, these attributes provide a comprehensive framework for the systematic study and management of alarms in RAN systems.

4. METHODOLOGY

This section provides a systematic approach to examining RAN alarm data, enabling the identification of key alarms and the assessment of their causal relationships, which are critical for improving network reliability and performance. Fig. 3 outlines the broad steps taken to develop the proposed "Alarm Webs" framework. The data was made available as a CSV file, which is curated using Python

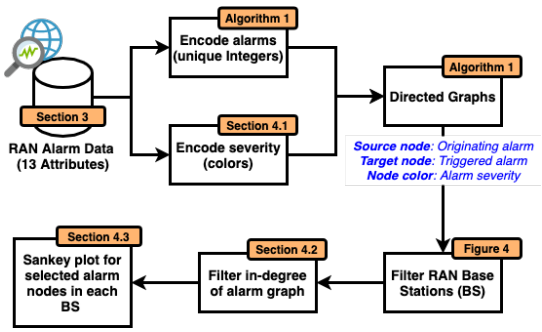


Fig. 3. Methodology followed in the proposed framework

Pandas library and the framework is developed using the Python 3 language on the Google Colaboratory platform.

Algorithm 1 Graph Construction for RAN Alarm Transitions

BS ← UniqueAlarms(df['Alarm Source'])
(Labels, UniqueAlarms) ← Factorize(df['Name'])
MappedLabels ← MapLabels(Labels, df['Name'])
AlarmMap ← CreateDataFrame('Alarm Name', 'Alarm Number', 'Severity')
AlarmMap ← RemoveDuplicates(AlarmMap)
GraphList ← InitializeList()
for $i \leftarrow 0$ **to** 100 **do**
    Graph ← CreateDiGraph()
    FilteredData ← Filter(df, BS[i])
    for all row ∈ FilteredData **do**
        if row['Next Alarm Number'] ≠ 999 **then**
            AddEdge(Graph, row['Alarm Number'], row['Next Alarm Number'])
        end if
    end for
    AddToList(GraphList, Graph)
end for

4.1 Data Encoding and Graph Construction

In this work, directed graphs (digraphs) are used to model RAN alarm systems and their severities. Algorithm 1 outlines the steps taken to construct the graph. Each alarm is encoded with a unique identifier, which serves as a node within the graph. A color-coded scheme visually represents alarm severities. Directed edges between nodes depict the alarm succession, where an edge from node A to B indicates A preceding B. The algorithm constructs the graph for RAN alarm transitions by first extracting unique alarm sources and mapping alarm names to labels. It then creates a DataFrame to store alarm information, removes duplicates, and initializes a list of graphs. The algorithm iteratively filters data for each unique alarm source, creating directed edges for sequential alarms, and adds the resulting graphs to a list for further analysis. Fig. 4 shows the graph for all the alarm inter dependencies in 6 randomly selected RAN base stations.

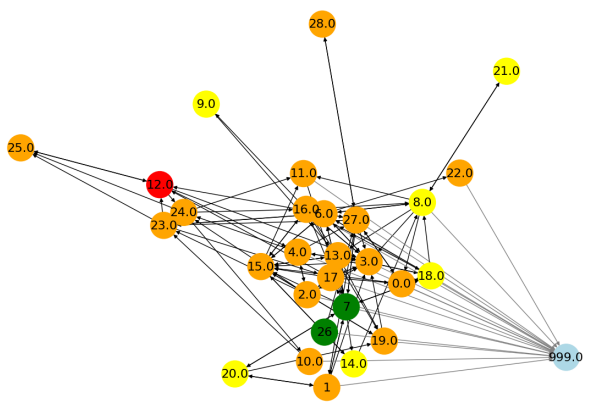
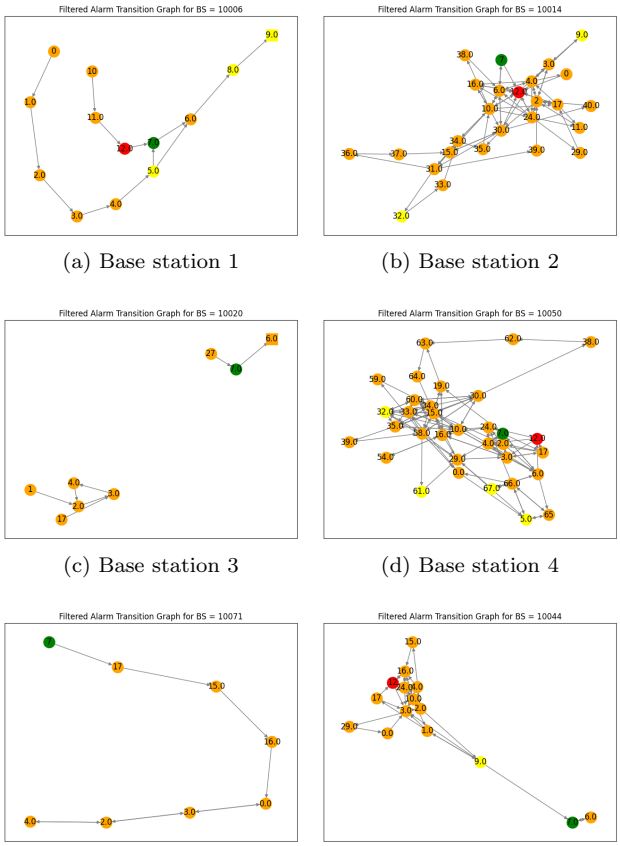


Fig. 4. A directed graph of the originating alarms and the subsequent alarms across 6 randomly selected RAN base stations



(a) Base station 1 (b) Base station 2

(c) Base station 3 (d) Base station 4

(e) Base station 5 (f) Base station 6

Fig. 5. Alarm graph dynamics for a few randomly selected RAN base stations

4.2 Degree Centrality Analysis

Algorithm 1 is modified to include filtering of RAN base stations. Now, for each base station under investigation, a directed graph is constructed from the alarm data (instead of the whole dataset). Within these graphs, the in-degrees and out-degrees of the nodes are calculated to determine the centrality of each alarm. Nodes with an in-degree greater than four are considered to be of particular interest, as this threshold suggests a higher frequency of occurrence following other alarms. These nodes are hypothesized to be indicative of systemic vulnerabilities or recurrent issues requiring further attention. Fig. 5 shows the various interdependencies discovered for various alarms within 6 randomly selected base stations. The observed variations in the alarm graph topologies between various base stations, relationships between the alarm severities and the ensuing relationships between the alarm themselves necessitates a robust approach for causality analysis. The first step to determining this causality between alarms is the visualization of these relationships (which we limit this work to) as described in the next section.

4.3 Causality Visualization & Interpretation

To unravel the causality and interdependencies among the alarms, Sankey diagrams are utilized. These diagrams are applied to the subset of nodes filtered by the degree centrality analysis. By focusing on nodes with high in-degrees,

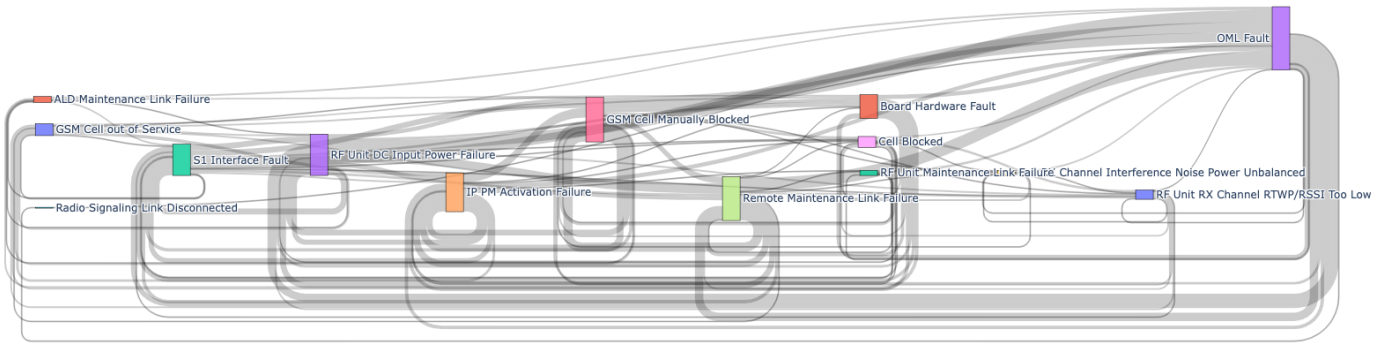


Fig. 6. Cause-effect relationship between the RAN alarms for alarm occurrences with in-degrees> 5

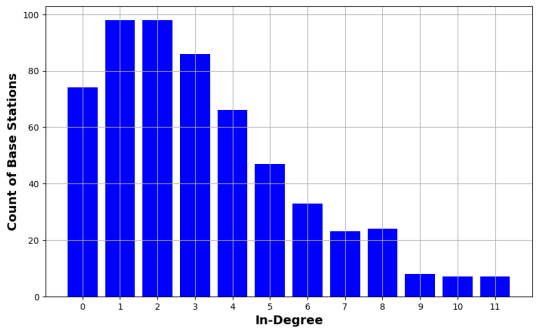


Fig. 7. A count of the in degrees of the various alarm nodes across 100 base stations

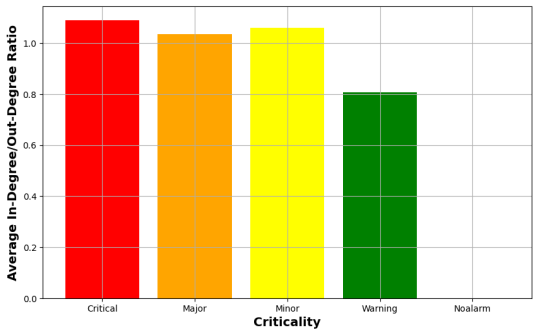


Fig. 8. Average in-degree to out-degree ratios of various alarm criticality across 100 base stations

the Sankey diagrams enable the easy visualization of the flow and magnitude of alarm sequences, thereby highlighting potential causal relationships between alarms. The width of the links in the Sankey diagrams corresponds to the frequency of the transitions, providing a quantitative measure of the relationships between alarms. Fig. 6 shows the visualization of the alarm nodes which have in-degrees greater than 5. By interpreting these diagrams, we aim to identify patterns and potential points of failure within the network. This analysis is expected to offer insights into the most critical alarms, which could inform maintenance strategies and contribute to the development of predictive models for alarm management.

5. ANALYSIS & DISCUSSION

Fig. 7 illustrates the distribution of in-degrees for alarm nodes across 100 RAN base stations. The x-axis represents the different in-degree values, while the y-axis shows the count of alarm nodes exhibiting each in-degree. Most base stations have alarm nodes with in-degrees of 1 and 2, indicating that one and two previous alarms led to this alarm, respectively. The distribution of the in-degrees is skewed, with a concentration of nodes having lower in-degrees, indicating that many alarms are not heavily connected to other alarms.

Alarm nodes within the in-degree range of 4 to 11 indicate notable connectivity and interdependence among alarms. This suggests a moderate to high number of incoming connections, signifying a cluster of alarms closely related within the network. Maintenance efforts should prioritize this cluster, considering potential systemic issues or shared dependencies among the alarms. Focusing on alarms with higher in-degrees, closer to 11, becomes crucial, as these nodes may significantly impact multiple components. This approach ensures that maintenance actions are strategically directed to enhance the overall resilience and stability of the base station, addressing potential cascading failures and contributing to effective troubleshooting efforts.

The Sankey diagram in Fig. 6 highlights the interdependence between alarm nodes with in-degrees exceeding 5 (the value is chosen for demonstration purposes in this paper) and offers a comprehensive understanding of the interconnections and dynamics within the network. The visual representation identifies central nodes with pivotal roles, showcasing their influence on multiple alarms. Clusters of interconnected alarms become apparent, guiding targeted maintenance efforts to specific areas of heightened connectivity. The diagram is a valuable tool for visualizing the flow of alarm propagation, interpreting patterns, and revealing potential weak points in the network. This combined approach enhances the ability to prioritize maintenance based on critical nodes. It facilitates a nuanced analysis of the network's resilience, empowering optimization strategies for a robust and reliable base station.

Further, Fig. 8 shows the in-degree to out-degree ratio for alarm criticalities and provides insights into the nature of alarms and how they propagate through the network. An alarm with a high in-degree (ratio > 1) signifies that multiple other alarms or events lead to this alarm. This information could indicate a critical or central issue in the network that is being triggered by various other faults or problems. Conversely, a high out-degree for an alarm (ratio < 1), indicates that the alarm leads to or causes several other alarms, which might be an initial

fault or trigger that sets off a cascade of other issues in the network. Alarm nodes with high in-degrees could be critical points in the network where issues converge, potentially indicating vital components that, when failing, have widespread effects, whereas alarm nodes with high out-degrees might be indicative of primary faults, whose resolution could prevent a series of subsequent issues.

Understanding the alarm relations and in-degree/out-degree ratios can aid in prioritizing which alarms should be addressed first. Alarms that lead to many others (high out-degree) might be tackled earlier to prevent cascading issues. Alarms that result from many causes (high in-degree) might require more comprehensive approaches to address underlying systemic issues. The ratio can help identify the root causes of widespread network issues. High-in-degree alarms might be symptoms of a deeper problem, while high-out-degree alarms might be closer to the root cause.

6. CONCLUSIONS

This research presents a preliminary framework to guide and prioritize maintenance efforts in network RAN infrastructures. The contributions of this research include the introduction of a novel digraph-based approach to alarm system analytics, developing a criterion using in-degree centrality for assessing alarm significance, and pioneering the use of Sankey diagrams in alarm sequence analysis. These contributions offer valuable insights for proactive maintenance and heightened reliability in communication base stations. This work addressed key research questions, exploring the role of graphs in identifying causality and assessing the influence of nodes in alarm systems across diverse base stations. The visualization of alarm sequences is investigated for its contribution to interpreting complex alarm data and predicting/preventing cascading failures. By visualizing alarms as graph nodes with causality-encoded edges, "Alarm Webs" enhances the understanding of alarm causality, prioritization, and complexity. The Sankey diagram further contributes to comprehending alarms' causality, priority, and complexity based on their connections and thickness.

Future work will include a more granular examination of alarm dynamics (focusing on intra-criticality alarm relationships), alarm priority determination through weighted assignments, exploring the geographical origin and distribution of alarms, and incorporating time series analysis. Approaches such as Dynamic Time Warping to compare time series data of alarms from different periods or base stations, survival analysis to predict the time until the next critical alarm or system failure, and Seasonal Decomposition of Time Series (SDTS) to understand how different times of the year impact the frequency and types of alarms would be interesting to explore.

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